

Maths Unit 3, Part 3A: Sets

Aasan Notes for Seemab · kitab ke page 46 se 52 tak · practice: Exercise 3.1 ke Q1 aur Q2

1. Set kya hai (p46)

 **Kitab ki definition, exam mein bilkul yehi English likhni hai**

"A set is a well-defined collection of distinct objects."

"The term well-defined means that the objects follow a given discipline with which presence or absence of some object in the set is checked."

Well-defined ka matlab: koi jhagra na rahe. "Halke patharon ka collection" set nahin hai, kyunke halka aur bhari har banda apni marzi se decide karega. "1 kg se kam wazan ke patharon ka collection" well-defined hai, kyunke tarazu faisla kar deta hai. Ek set mein alag alag qism ki cheezein bhi aa sakti hain, jaise library ke books, chairs, keys aur locks.

Example 1 (p46)

Prime aur even dono = $\{2\}$ · Pakistani currency = $\{5, 10, 20, 50, 100, 500, 1000, 5000\}$ · lafz 'set' ke huroof = $\{s, e, t\}$ · 1 se chhote natural numbers = $\{1\}$ · $-3, -9, 2, 3, 4, 2$ mein se natural numbers = $\{2, 3, 4\}$. Aakhri wala yaad rakho: repeat honay wala element sirf **ek dafa** likha jata hai.

2. Set-builder form (rule method) (p47)

Notation aur uski reading

Yahan saray elements ginte nahin, balke un ka **rule** likhte hain. $A = \{x \mid x \in Z\}$ ko parha jata hai: **"A is the set of all elements x such that x belongs to Z"**. Seedhi lakeer $\mid$ = "such that", aur $\wedge$ = "and".

Example 2 (p47): lafzon se set-builder banao

B = Prime numbers less than 17

$B = \{x \mid x \in P \wedge x < 17\}$

C = multiples of 4 greater than or equal to 40

$C = \{4x \mid x \in N \wedge x \geq 10\}$

D = $\{1, 2, 3, 6\}$

$D = \{x \mid x \text{ is a factor of } 6\}$

E = squares of 1st three natural multiples of 10

$E = \{(10x)^2 \mid x \in N \wedge 1 \leq x \leq 3\}$

Example 3 (p47, p48): set-builder se descriptive ya tabular

$G = \{15x \mid x \in Z \wedge x \geq 1\}$

Set of integral multiples of 15, greater than or equal to 15

$H = \{y \mid y \in W \wedge y^3 - 1 = 7\}$

$H = \{2\}$

$I = \{x \mid x \in W \wedge -3 > x > -5\}$

$I = \{\}$

$J = \{x/2 \mid x \in E \wedge x > 0\}$

$J = \{1, 2, 3, 4, \dots\}$

$K = \{y \mid y \in Z^- \wedge y^2 = 9\}$

$K =$ Set of negative integers containing '−3' only

Pakki trick

Multiples ka sawal aaye to element ki jagah **4x** ya **15x** likho aur shart x par lagao. "40 se shuru" = $4x \geq 40$, yaani **x** ≥ 10 , na ke $x \geq 40$.

3. Most commonly used sets of numbers (p48)

N = Natural numbers

$\{1, 2, 3, 4, \dots\}$

P = Prime numbers

$\{2, 3, 5, 7, 11, 13, \dots\}$

W = Whole numbers

$\{0, 1, 2, 3, 4, \dots\}$

Q = Rational numbers

$\{p/q \mid p, q \in Z \wedge q \neq 0\}$

Z = Integers

$\{0, \pm 1, \pm 2, \pm 3, \dots\}$

Q' = Irrational numbers

$\{x \mid x \neq p/q, p, q \in Z \wedge q \neq 0\}$

E = Even numbers

$\{0, \pm 2, \pm 4, \pm 6, \dots\}$

R = Real numbers

$\{x \mid x \in Q \vee x \in Q'\}$

O = Odd numbers

$\{\pm 1, \pm 3, \pm 5, \dots\}$

N se O tak pehle descriptive phir tabular (roster) form; Q, Q' aur R pehle descriptive phir **set-builder** form. Check Point: R tabular form mein nahin likh sakte, real numbers gine nahin ja sakte.

4. Set operations (p48, p49)

Operation	Kitab ki definition	Set-builder	Kitab ki misaal
Union $A \cup B$	Woh elements jo ya A se, ya B se, ya dono se liye jayen	$\{x \mid x \in A \vee x \in B\}$	$A = \{1, 2, 3, 4, 5\}, B = \{3, 4, 1, 7, 6\}$ $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$
Intersection $A \cap B$	Sirf woh elements jo A aur B dono mein hon	$\{x \mid x \in A \wedge x \in B\}$	Wohi A aur B $A \cap B = \{1, 3, 4\}$
Difference $A - B$	A ke woh elements jo B mein maujood nahin	$\{x \mid x \in A \wedge x \notin B\}$	$A = \{1, 2, 3, 4, 5\}, B = \{3, 4, 6, 7, 8\}$ $A - B = \{1, 2, 5\}$ Yaad rakho $A - B \neq B - A$, kyunke $B - A = \{6, 7, 8\}$

Operation	Kitab ki definition	Set-builder	Kitab ki misaal
Complement A^c ya A'	U ke woh saray elements jo A mein nahin ($A \subset U$ hona zaroori)	$A' = U - A$ $= \{x \mid x \in U \wedge x \notin A\}$	$U = \{1, \dots, 7\}$, $A = \{2, 4, 6\}$ $A^c = \{1, 3, 5, 7\}$

★ Key Fact box (p49), yahan se MCQ banti hai

- The common elements of A and B are written once.
- Two sets A and B are said to be disjoint, if $A \cap B = \varnothing$
- The elements of A are never present in A^c and vice versa.
- Two sets are said to be overlapping, if neither set is subset of the other and their intersection is non-empty.

5. Do sets ka rishta aur Venn diagram (p49, p50)

Rishta	Shart	Kitab ki misaal aur uska Venn
Disjoint	Koi element common nahin, yaani $A \cap B = \varnothing$	$A = \{1, 2, 3\}$, $B = \{4, 5\}$. Do alag circles jo na chhuein.
Overlapping	(i) kam az kam ek element common ho, aur (ii) koi set doosre ka subset na ho	$A = \{2, 3, 5, 7, 9\}$, $B = \{0, 3, 5, 8\}$, $A \cap B = \{3, 5\} \neq \varnothing$. Do circles jo beech mein kat-te hain.
Subset $A \subset B$	Set A ka har element set B mein bhi maujood ho	$A = \{a, b, c\}$, $B = \{a, b, c, d, \dots\}$. Chhota circle bare ke bilkul andar.

📖 Venn diagram banane ka tareeqa (p50)

"The pictorial representation of the relationship $A \subset U$ is called a Venn diagram. In a Venn diagram, universal set is represented by the interior of the rectangle, however inside the rectangle, the subsets are represented by the interior of any other closed shape like circles or ovals etc."

3 6. Teen sets ka union aur intersection (p50, p51, p52)

🧠 Bracket ka usool, sab se ahem baat

Kitab: "We use brackets to separate two sets from the third one because these represent different sets." Pehle **bracket ke andar** wala kaam karo, phir us ka teesre set ke saath. Is liye **$(A \cup B) \cap C$** aur **$A \cup (B \cap C)$** bilkul alag cheezein hain.

🔢 Example 5 (p51): $A = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{6, 7, 8, 9\}$

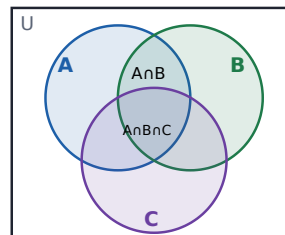
$B \cap C = \{6\}$, to **$A \cup (B \cap C) = \{0, 1, 2, 3, 4, 6\}$** . $B \cup C = \{3, 4, 5, 6, 7, 8, 9\}$, to **$A \cap (B \cup C) = \{3, 4\}$** .

🔢 Example 6 (p51), disjoint ka case: $P = \{a, b, c, d, e\}$, $Q = \{f, g, h, i\}$, $R = \{d, e, f, g, j\}$

$P \cup Q = \{a, b, c, d, e, f, g, h, i\}$, to **$(P \cup Q) \cap R = \{d, e, f, g\}$** . P aur Q disjoint hain ($P \cap Q = \varnothing$), to **$(P \cap Q) \cup R = \{d, e, f, g, j\}$** .

🔢 Example 7 (p51, p52), subset ka case: $P = \{3, 5, 7\}$, $Q = \{1, 3, 5, 7, 9\}$, $R = \{0, 2, 5, 8\}$

$P \subset Q$ hai, is liye P ka circle Q ke andar banega. $Q \cap R = \{5\}$, to **$P \cup (Q \cap R) = \{3, 5, 7\}$** .



Teen overlapping sets ka standard Venn. Rectangle = U , beech ka chhota hissa $A \cap B \cap C$. Shade karne se pehle poora naqsha banao, phir maanga gaya hissa bharo.

Example 4 (p50, p51) inhi A, B, C ka teen-circle Venn hai. Ab **Exercise 3.1 ke Q1 aur Q2** khud karo.

🔪 Khud ko check karo: 5 sawal

- Set-builder form mein likho:** $C =$ Set of multiples of 7 greater than or equal to 35.
- Nikalo.** $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 6, 7, 8\}$. Batao: (i) $A \cup B$ (ii) $A \cap B$ (iii) $A - B$ (iv) A^c .
- Nikalo.** $A = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{6, 7, 8, 9\}$. Batao (i) $(A \cup B) \cap C$ (ii) $A \cap (B \cup C)$. Dono jawab alag kyun?
- Venn banao aur shade karo.** $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$, $C = \{5, 6\}$. Teen circles ka Venn banao, $(A \cap B) \cup C$ shade karo, aur batao A aur C ka rishta kya hai (disjoint, overlapping ya subset).
- Define karo** (kitab ki English mein): a set, aur "well-defined". Phir batao ke "a collection of lighter stones" set hai ya nahin, aur kyun.

✅ Jawab

- $C = \{7x \mid x \in \mathbb{N} \wedge x \geq 5\}$, kyunke $7x \geq 35$ ka matlab $x \geq 5$ hai.
- (i) $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$ (ii) $A \cap B = \{3, 4\}$ (iii) $A - B = \{1, 2, 5\}$ (iv) $A^c = U - A = \{6, 7, 8\}$
- $A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$, to **$(A \cup B) \cap C = \{6\}$** . $B \cup C = \{3, 4, 5, 6, 7, 8, 9\}$, to **$A \cap (B \cup C) = \{3, 4\}$** . Jawab alag hain kyunke bracket pehle chalta hai, to kaam ki tarteeb badal jati hai.
- $A \cap B = \{3, 4\}$, $C = \{5, 6\}$, to $(A \cap B) \cup C = \{3, 4, 5, 6\}$. Venn mein **A aur B ka common hissa** aur **poora C ka circle** shade honge. $A \cap C = \varnothing$, is liye A aur C **disjoint** hain.
- "A set is a well-defined collection of distinct objects." "The term well-defined means that the objects follow a given discipline with which presence or absence of some object in the set is checked." Lighter stones wala collection **set nahin**, "halka" ki koi tay shuda hadd nahin.